

2.3 Probability

Mathematics B (MEI) — OCR B — A-Level (H640) — Statistics — spec refs u1–u7

What this topic covers. Calculating probabilities in a finite sample space, mutually exclusive and independent events, the addition and multiplication rules, Venn and tree diagrams, and conditional probability including the reverse case.

1. Basic probability

For equally likely outcomes,

$$P(A) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

Every probability satisfies $0 \leq P(A) \leq 1$. An impossible event has probability 0; a certain event has probability 1.

Complementary events

A' is the event "not A ", and

$$P(A') = 1 - P(A)$$

For "at least one" questions, the complement is almost always faster.

$P(\text{at least one six in two rolls}) = 1 - P(\text{no sixes}) = 1 - \left(\frac{5}{6}\right)^2 = \frac{11}{36}$, which is far quicker than listing the cases.

Expected frequency

$$\text{expected frequency} = n P(A)$$

In 200 trials of an event with probability 0.15, the expected frequency is $200 \times 0.15 = 30$.

2. Notation and diagrams

Notation	Meaning
$A \cap B$	A and B both occur
$A \cup B$	A or B (or both) occurs
A'	A does not occur
$P(A \mid B)$	The probability of A given that B has occurred

Three diagram types support the calculations: **tree diagrams** for successive events, **Venn diagrams** for overlapping events (up to three, in this specification), and **sample space** or **two-way tables** for enumerating outcomes.

3. Mutually exclusive events

Two events are **mutually exclusive** if they cannot both occur:

$$P(A \cap B) = 0, \quad P(A \cup B) = P(A) + P(B)$$

4. Independent events

Two events are **independent** if the occurrence of one does not affect the probability of the other:

$$P(A \cap B) = P(A) P(B)$$

Equivalently, $P(A \mid B) = P(A)$ (spec ref u7).

Mutually exclusive and independent are **different, and generally incompatible**. If A and B are mutually exclusive with non-zero probabilities, then knowing B happened tells you A definitely did not — so they cannot be independent. Assuming that "can't happen together" implies independence is a serious conceptual error.

Worked example. A machine has three independent components, each working with probability 0.9.

$$P(\text{all three work}) = 0.9^3 = 0.729$$

$$P(\text{at least one fails}) = 1 - 0.729 = 0.271$$

5. The general addition rule

For *any* two events ($u5$):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The subtraction avoids double-counting the overlap.

Worked example. $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.2$.

$$P(A \cup B) = 0.4 + 0.5 - 0.2 = 0.7$$

Worked example. Drawing one card from a standard pack, find $P(\text{king or heart})$.

$$\frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52}$$

The king of hearts is subtracted because it belongs to both.

6. Conditional probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Rearranged, this gives the multiplication rule $P(A \cap B) = P(B)P(A \mid B)$, which is exactly what a tree diagram encodes: multiply along the branches.

Worked example. $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cup B) = 0.7$. Find $P(A \mid B)$.

First $P(A \cap B) = 0.5 + 0.4 - 0.7 = 0.2$, so

$$P(A \mid B) = \frac{0.2}{0.4} = 0.5$$

Worked example. A bag holds 5 red and 3 blue balls. Two are drawn without replacement. Find $P(\text{exactly one red})$.

Two orders are possible:

$$P(RB) = \frac{5}{8} \times \frac{3}{7} = \frac{15}{56}, \quad P(BR) = \frac{3}{8} \times \frac{5}{7} = \frac{15}{56}$$

$$P(\text{exactly one red}) = \frac{30}{56} = \frac{15}{28}$$

"Without replacement" changes the denominator at the second draw, and often the numerator too — the events are **not** independent. "With replacement" leaves both unchanged.

Reverse conditional probability

Given $P(B \mid A)$ and asked for $P(A \mid B)$, work from the definition (u6). Building a tree diagram and totalling the relevant branches is usually the cleanest route.

Worked example. A screening test is 95% accurate in both directions, and 1% of the population has the condition. A person tests positive — what is the probability they have it?

Two ways to test positive:

$$P(\text{condition and positive}) = 0.01 \times 0.95 = 0.0095$$

$$P(\text{no condition and positive}) = 0.99 \times 0.05 = 0.0495$$

$$P(\text{positive}) = 0.0095 + 0.0495 = 0.059$$

$$P(\text{condition} \mid \text{positive}) = \frac{0.0095}{0.059} \approx 0.161$$

That result is worth pausing on: despite a 95% accurate test, roughly **84% of positives are false**. Because the condition is rare, the many small false-positive rates among the healthy majority outweigh the true positives. This is the classic reason a conditional probability cannot be reversed by intuition.

Conditional probability from a restricted sample space

Worked example. Three fair coins are tossed. Given that at least one head appears, find $P(\text{all three heads})$.

$P(\text{all heads}) = \frac{1}{8}$ and $P(\text{at least one head}) = 1 - \frac{1}{8} = \frac{7}{8}$, so

$$P(\text{all three} \mid \text{at least one}) = \frac{1/8}{7/8} = \frac{1}{7}$$

7. Common pitfalls

- Adding probabilities for events that are not mutually exclusive, forgetting to subtract the overlap.
- Multiplying probabilities for events that are not independent.
- Treating "mutually exclusive" as though it meant "independent".
- Keeping the same denominator on the second draw of a without-replacement problem.
- Reversing a conditional probability by intuition rather than by calculation.
- Forgetting that all branches from a node on a tree diagram must sum to 1.
- Giving a probability outside $[0, 1]$ and not noticing.